DS805

DIABETES RISK PREDICTION

CONTENTS

INTRODUCTION

- The Clinical Context: Diabetes is a growing, severe chronic health crisis.
- The Screening Bottleneck: Traditional screening relies on blood tests (fasting glucose, A1C), which require clinic visits, time, and money.
- Our Approach: Can we predict a patient's diabetes risk using only easily answered survey questions about their lifestyle, age, and basic health history?

BUSINESS PROBLEM

Organization: Regional public health department / Healthcare provider.

The Challenge: Missing an at-risk patient leads to delayed diagnosis and preventable complications (heart disease, neuropathy).

Goal: Proactively identify patients at risk of prediabetes/diabetes for early intervention programs.

The Trade-off: A false alarm (a benign follow-up call) is highly preferable to missing a diabetic patient.

BUSINESS METRIC - AT RISK CALL

Primary Metric

At-Risk Recall (Sensitivity for class 0).

The Cost Imbalance

The cost of a False Negative (missing a diabetic patient) is far greater than the cost of a False Positive (a benign follow-up call).

Target Goal

Maximize Sensitivity for Class 0, even if it slightly reduces overall accuracy.

DATASET OVERVIEW & STRATIFIED SAMPLING

1. **SOURCE**
   CDC BRFSS 2015 Survey

2. **TARGET (Y)**
   0 = At-risk, 1 = Healthy

3. **CHALLENGE**
   Severe class imbalance & massive scale (253,680 rows)

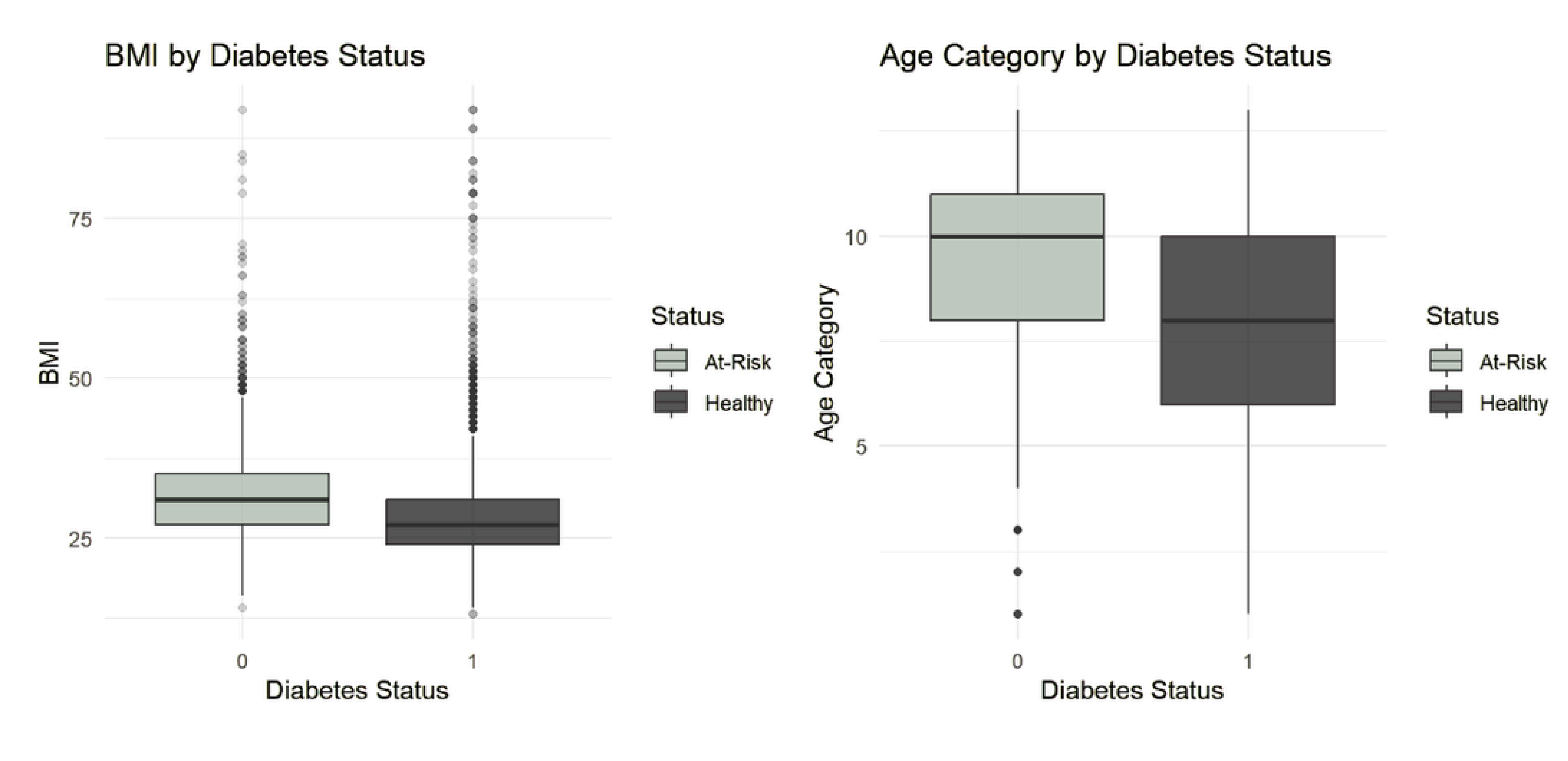
4. **SOLUTION**
   Stratified sampling down to 15,000 rows to ensure distance-based models can run efficiently

KEY PREDICTORS (DATA DICTIONARY)

- Features: 21 self-reported health indicators. No clinical lab tests are required.
- Numeric Variables: BMI, Age, GenHlth, MentHlth, PhysHlth, Education, Income.
- Categorical Variables: HighBP, HighChol, Smoker, Stroke, DiffWalk, etc.

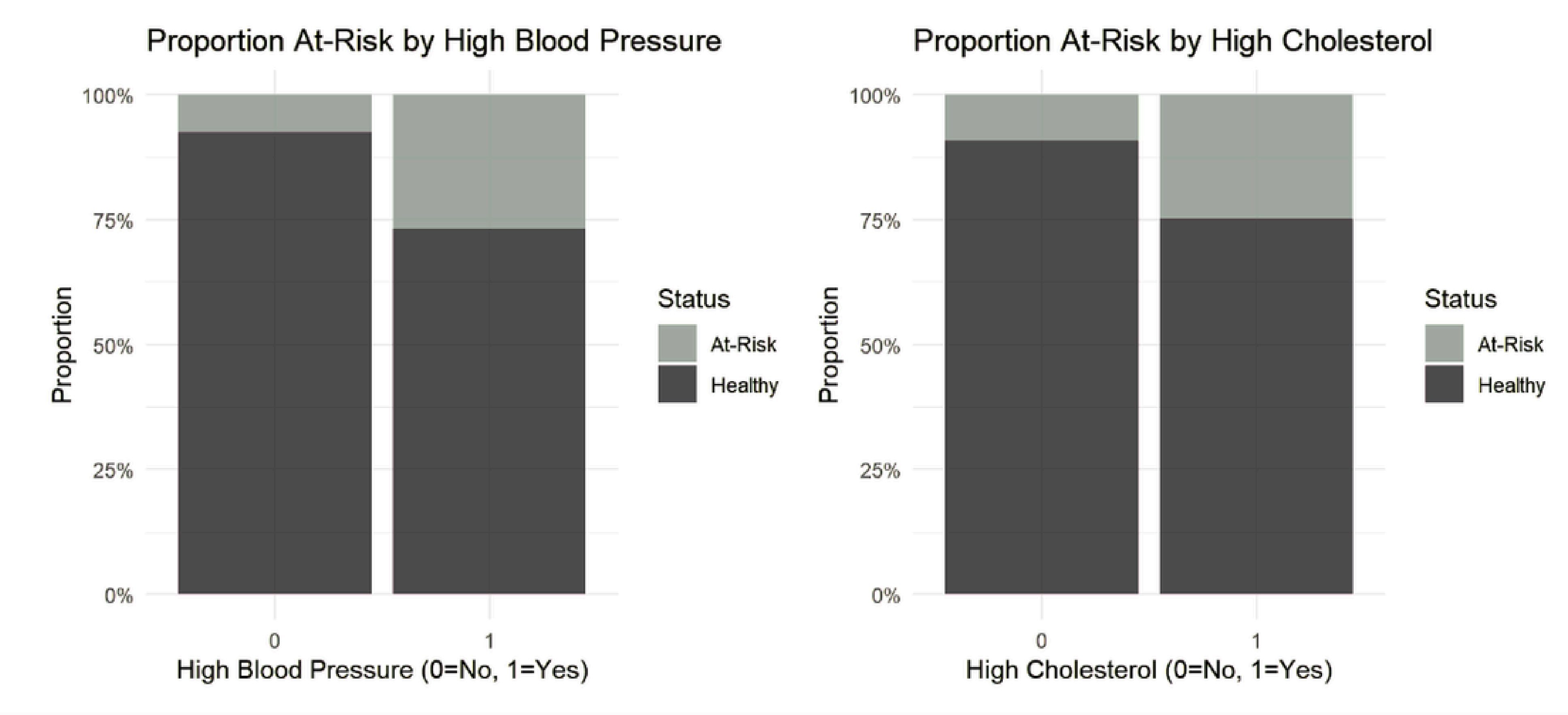
Variable	Type	Description
y	Factor (target)	0 = at-risk (prediabetes/diabetes), 1 = healthy
HighBP	Factor (0/1)	Ever told you have high blood pressure?
HighChol	Factor (0/1)	Ever told you have high cholesterol?
CholCheck	Factor (0/1)	Cholesterol check in last 5 years?
BMI	Numeric	Body mass index
Smoker	Factor (0/1)	Have you smoked ≥100 cigarettes in your life?
Stroke	Factor (0/1)	Ever told you had a stroke?
HeartDiseaseorAttack	Factor (0/1)	Ever told you have CHD or MI?

EDA - DEMOGRAPHICS



- BMI: At-risk median is ~31 (Obese) vs. Healthy median ~27 (Overweight).
- Age: At-risk individuals are generally 2-3 age brackets (10-15 years) older than the healthy group.

EDA - COMORBIDITY PATTERNS



- High Blood Pressure: Roughly 2x higher risk of diabetes compared to baseline.
- Heart Disease: The strongest single separator (nearly 40% of patients with heart disease are at-risk for diabetes).

TRAIN-TEST SPLIT

- Split Strategy: 80/20 Stratified Split.
- Goal: Guarantee that the exact 15.7% proportion of at-risk patients is identical in both the training and testing sets.

dataset	n
train	12002
test	2999

THE BASELINE FAILURE

```
          Reference
Prediction    0    1
         0    0    0
         1  472 2527

               Accuracy : 0.8426
                 95% CI : (0.8291, 0.8555)
    No Information Rate : 0.8426
    P-Value [Acc > NIR] : 0.5123

 Mcnemar's Test P-Value : <2e-16

            Sensitivity : 0.0000
            Specificity : 1.0000
         Pos Pred Value :    NaN
         Neg Pred Value : 0.8426
             Prevalence : 0.1574
         Detection Rate : 0.0000
   Detection Prevalence : 0.0000
      Balanced Accuracy : 0.5000
```

Predict "Healthy" (the majority class) for every single patient.

STRATEGY

High (84.26%)

ACCURACY

0.00%

AT-RISK RECALL (SENSITIVITY)

LOGISTIC REGRESSION (STANDARD VS. TUNED THRESHOLD)

threshold	test_error	at_risk_recall
0.5	0.1601	0.1547

threshold	test_error	at_risk_recall
0.7	0.1807	0.4576

MODEL	THRESHOLD 0.50	THRESHOLD 0.70
Regression using all 21 predictors.	Strong overall accuracy but catches very few sick patients	Requires a 70% probability of being healthy to avoid a flag. This drastically improves recall

LOGISTIC REGRESSION INSIGHTS

```
                      Estimate Std. Error z value Pr(>|z|)    
(Intercept)           7.000125   0.388202  18.032  < 2e-16 ***
HighBP1              -0.609711   0.063005  -9.677  < 2e-16 ***
HighChol1            -0.609995   0.059132 -10.316  < 2e-16 ***
CholCheck1           -1.137697   0.275016  -4.137 3.52e-05 ***
BMI                  -0.064972   0.004039 -16.085  < 2e-16 ***
Smoker1               0.010515   0.057886   0.182  0.85585    
Stroke1               0.001924   0.116317   0.017  0.98680    
HeartDiseaseorAttack1 -0.123191  0.080482  -1.531  0.12585    
PhysActivity1         0.174150   0.062781   2.774  0.00554 ** 
Fruits1               0.001000   0.060141   0.017  0.98673    
Veggies1              0.046181   0.070262   0.657  0.51101    
HvyAlcoholConsump1    0.633276   0.160156   3.954 7.68e-05 ***
AnyHealthcare1        0.042404   0.136696   0.310  0.75640    
NoDocbcCost1         -0.056728   0.102000  -0.556  0.57810    
GenHlth              -0.430307   0.034920 -12.323  < 2e-16 ***
MentHlth             -0.002092   0.003695  -0.566  0.57129    
```

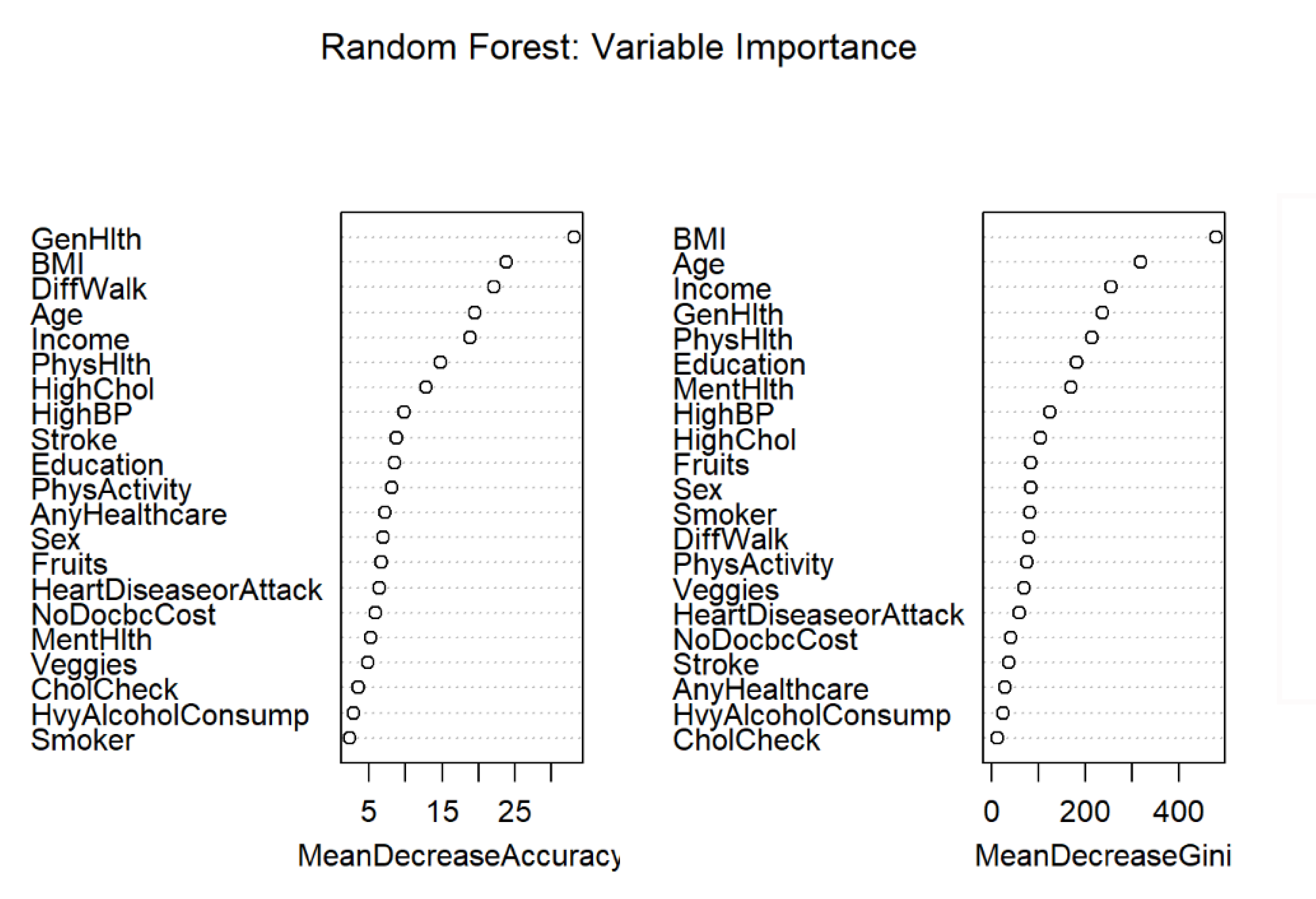
```
MentHlth             -0.002092   0.003695  -0.566  0.57129    
PhysHlth              0.008906   0.003489   2.553  0.01068 *  
DiffWalk1            -0.119747   0.075193  -1.593  0.11126    
Sex1                 -0.328182   0.058914  -5.570 2.54e-08 ***
Age                  -0.136079   0.012232 -11.125  < 2e-16 ***
Education             0.047456   0.030567   1.553  0.12054    
Income                0.079780   0.015549   5.131 2.88e-07 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 10459.4  on 12001  degrees of freedom
Residual deviance:  8372.5  on 11980  degrees of freedom
AIC: 8416.5

Number of Fisher Scoring iterations: 6
```

K-NEAREST NEIGHBORS (KNN)

k	Error_Rate	AtRisk_Recall
3	0.1804	0.2352
5	0.1727	0.1886
7	0.1694	0.1568
11	0.1671	0.1229
15	0.1634	0.1102

- Data Prep: Scaled all features to accurately calculate Euclidean distance.
- Tuning: Evaluated multiple 'k' neighborhood sizes (3, 5, 7, 11, 15).
- Limitation: Imbalanced data causes the healthy majority to constantly outvote the sick minority in local neighborhoods.

RANDOM FOREST - VARIABLE IMPORTANCE



- Top 5 Predictors: General Health, BMI, Age, High Blood Pressure, and High Cholesterol.
- Insight: Self-rated subjective health is the single most powerful node-splitter in the forest.

RANDOM FOREST - FINAL PERFORMANCE

```
Confusion Matrix and Statistics

          Reference
Prediction    0    1
         0  233  325
         1  239 2202

               Accuracy : 0.8119
                 95% CI : (0.7975, 0.8258)
    No Information Rate : 0.8426
    P-Value [Acc > NIR] : 0.9999971

                  Kappa : 0.3399

 Mcnemar's Test P-Value : 0.0003447

            Sensitivity : 0.49364
            Specificity : 0.87139
         Pos Pred Value : 0.41756
         Neg Pred Value : 0.90209
             Prevalence : 0.15739
         Detection Rate : 0.07769
   Detection Prevalence : 0.18606
      Balanced Accuracy : 0.68252

       'Positive' Class : 0
```

- Performance (Threshold 0.70): Achieves an At-Risk Recall of 49.36%.
- Business Impact: We successfully catch almost half of all diabetic patients using nothing but survey data.

SUPPORT VECTOR MACHINE (SVM)

```
Confusion Matrix and Statistics

          Reference
Prediction    0    1
         0   39   31
         1  433 2496

               Accuracy : 0.8453
                 95% CI : (0.8318, 0.858)
    No Information Rate : 0.8426
    P-Value [Acc > NIR] : 0.3553

                  Kappa : 0.1076

 Mcnemar's Test P-Value : <2e-16

            Sensitivity : 0.08263
            Specificity : 0.98773
         Pos Pred Value : 0.55714
         Neg Pred Value : 0.85217
             Prevalence : 0.15739
         Detection Rate : 0.01300
   Detection Prevalence : 0.02334
      Balanced Accuracy : 0.53518

       'Positive' Class : 0
```

- Kernel: Radial Basis Function (RBF) for non-linear decision boundaries.
- Tuning: 5-fold Cross-Validation utilized to find optimal cost and gamma.
- Result: Yields the highest overall accuracy (84.5%), but terrible recall (8.2%).

ERROR ANALYSIS & MODEL COMPARISON

Model	Tuning	Test_Accuracy	Test_Error	AtRisk_Recall	Specificity
Majority Baseline	None	0.8426	0.1574	0.0000	1.0000
Logistic Regression (t=0.50)	All predictors; threshold=0.50	0.8399	0.1601	0.1547	0.9679
Logistic Regression (t=0.70)	All predictors; threshold=0.70	0.8193	0.1807	0.4576	0.8868
KNN (k=3)	k=3; scaled features	0.8203	0.1797	0.2352	0.9296
Random Forest (t=0.70)	ntree=300, mtry=4; threshold=0.70	0.8119	0.1881	0.4936	0.8714
SVM (Radial, default t=0.50)	cost=1, gamma=0.1	0.8453	0.1547	0.0826	0.9877

Model Performance by Age Segment (Random Forest, t=0.70)

Age_Group	n	n_atrisk	Accuracy	Error_Rate	AtRisk_Recall
Younger (18-39)	431	10	0.9652	0.0348	0.3000
Middle (40-59)	1131	136	0.8621	0.1379	0.4853
Older (60+)	1437	326	0.7265	0.2735	0.5031

CONCLUSION
RECOMMENDATIONS & FUTURE WORK

- **Deployment Action:** Deploy the Random Forest model (t=0.70) to automatically flag patients below a 0.70 healthy probability for care coordinator outreach.
- **Monitoring:** Track data drift as survey population demographics shift over time.
- **Future Work:** Build age-stratified models (specifically for the 18-39 bracket) and incorporate at least one objective clinical metric, such as a fasting glucose test.

THANK

YOU:)